



JSP I

1. JSP as complementary technology
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5. Implicit objects

- JSP(Java Server Pages) is built on top of servlets.
- JSP pages are simply HTML pages with embedded Java.
- JSP (JavaServer Pages) and servlets serve different purposes and can be effectively used together as complementary technologies.
- Servlets used for Handle core logic whereas JSP used for Combines static HTML markup with Java code snippets (scriptlets) for data and logic.
- Difference between servlets and JSP.

- Several key stages in JSP life cycle
 1. Translation: JSP page converted into a Java servlet class
 2. Compilation: The generated servlet class is then compiled into a Java bytecode
 3. Class Loading: The compiled bytecode class is loaded into web server's memory
 4. Instantiation: Generated servlet class is created
 5. Initialization: `jspInit()` method of the generated servlet is called
 6. Request Processing: `_jspService()` method of the generated servlet is called
 7. Destruction: `jspDestroy()` method of the generated servlet is called
 8. Unloading: the web server may unload the generated servlet class from its memory

- JSP elements are the building blocks of JSP pages.
- During translation, the JSP engine converts these elements into equivalent Java code
- JSP is basically HTML and additional JSP elements like
 1. Scriptlets (`<% %>`)
 2. Expressions (`<%= %>`)
 3. Directives (`<%@ %>`)
 4. Declarations (`<%! %>`)
 5. JSP Actions (`<jsp:action>`)
 6. JSP Standard Tag Library (JSTL) Tags
 7. Custom Tags
 8. Comments (`<%-- --%>`)

4. Creating simple JSP

```
<%@ page import="java.util.*" language="java" ....%>
<html>
<body>
  <h1> Welcome to JSP </h1>
  This is template text
  <!-- This is HTML comment -->
  <%-- This is JSP comment --%>
  <%-- Scriptlet --%>
  <%
    int n = 0;
    out.print("Value of n : "+(++n));
  %>
  <% out.print(Calendar.getInstance().getTime()); %>
  <%-- Expression %>
  <%= Calendar.getInstance().getTime() %>
  <%-- Declaration -->
  <%! int iVar; %>
  <%= "value of iVar : "+(++iVar) %>
</body>
</html>
```

- JSP container automatically creates implicit objects and makes available to all JSP pages.
- These objects provide access to various information.
- Common implicit objects are:
 1. request
 2. response
 3. session
 4. application
 5. config
 6. pageContext
 7. page
 8. exception